Supporting Information

MolScribe: Robust Molecular Structure Recognition with Image-To-Graph Generation

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Experiment Details

Our experiments are conducted on a Linux server with 96 CPUs and 500GB RAM. We use four NVIDIA A100 GPUs to train our models. Our neural network implementation is based on PyTorch.[1]

Our code can be found at `https://github.com/thomas0809/MolScribe`, which includes a Python interface for using MolScribe to make predictions, and scripts for training and evaluating MolScribe. Our web interface (`https://huggingface.co/spaces/yujieq/MolScribe`) is implemented in Hugging Face as a Gradio space.[2]

Data

Training The synthetic training data is constructed with molecules sampled from PubChem (`https://ftp.ncbi.nlm.nih.gov/pubchem/Compound/`) and images automatically

rendered by Indigo. Unlike previous work, we do not apply any filtering based on molecule size or other properties, but randomly sample the molecules. We make a few changes to the source code of Indigo to support more rendering options and obtain pixel-level atom coordinates. The modified code is included in our codebase.

The patent training data is downloaded from USPTO, Grant Red Book section (`https://bulkdata.uspto.gov/`). We collect the MOLfiles and corresponding images, and use RDKit to convert molecules into SMILES strings. Relative coordinates, bond types, and superatom labels (functional group abbreviations, R-groups) are parsed from the MOLfiles.

The patent data is noisier than the synthetic data. Figure 1 shows four common types of noise in the patent data. First, exact atom coordinates in the image are not available, but relative coordinates can be found in the MOLfiles. We scale the relative coordinates to the size of the image as shown in Figure 1 (a). In other words, the coordinates are translated and scaled, such that the left-most atom lies on the left edge of the image, the right-most atom lies on the right edge of the image, etc. The scaled coordinates are often shifted from the actual coordinates in the image. Second, the abbreviated functional groups are sometimes expanded in the MOLfile, but our formulation expects them to be condensed as superatoms. Third, the ground truth layouts in the MOLfiles do not match the images in some (but not many) cases, and some substructures can be wrong. Fourth, there may be extra annotations of text labels or lines. Despite the noise, we still find it helpful to include the patent data as part of our training data. We encourage future work to clean up this data and reduce the noise in training.

Testing The synthetic image benchmarks are automatically generated by two chemistry toolkits, Indigo and ChemDraw. We use the same set of molecules as the USPTO benchmark mentioned below.[3]

The realistic image benchmarks are taken from Rajan et al. and Staker et al.. CLEF, JPO, UOB, and USPTO are downloaded from `https://github.com/Kohulan/OCSR_Revi`

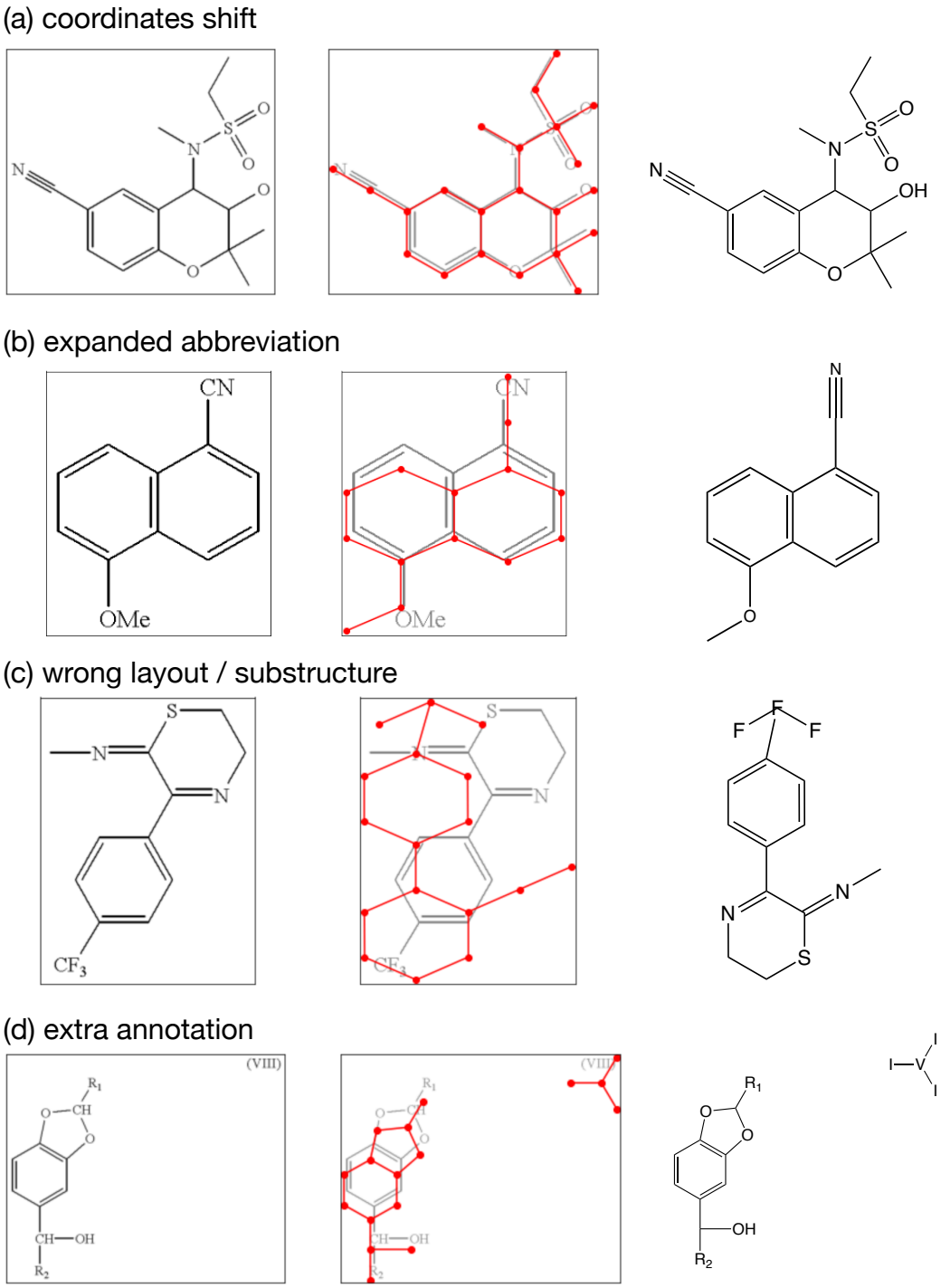


Figure 1: Common types of noise in the patent training data (USPTO). Left: molecular images. Middle: red dots represent the normalized coordinates of the atoms. Right: ground truth molecular structures.

`ew`, and Staker is downloaded from `https://drive.google.com/drive/folders/16OjPwQ7bQ486VhdX4DWpfYzRsTGgJkSu`. The perturbed datasets are constructed by Clevert et al.

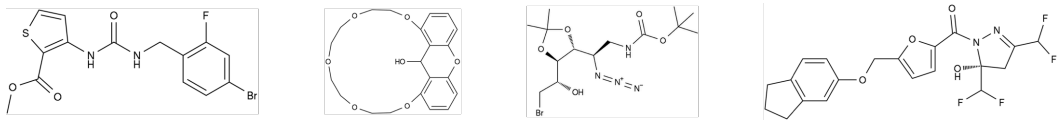
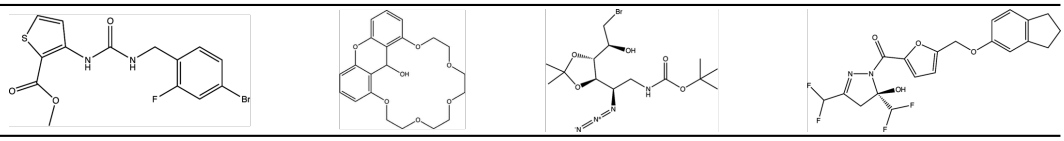
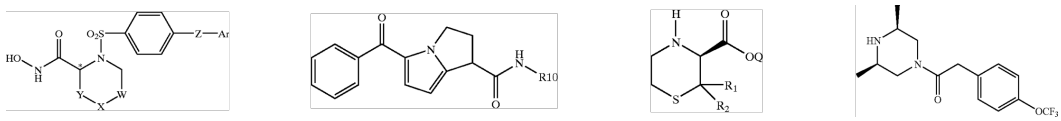
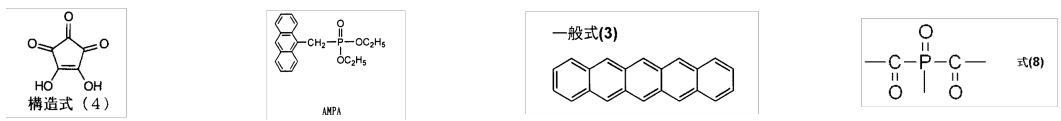
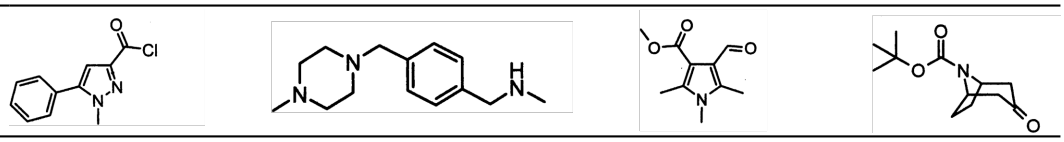
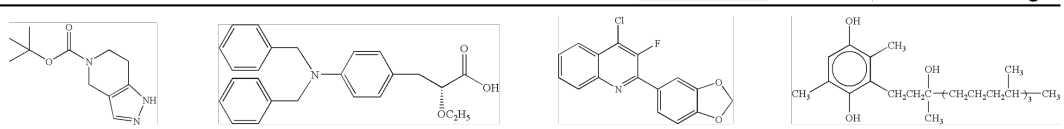
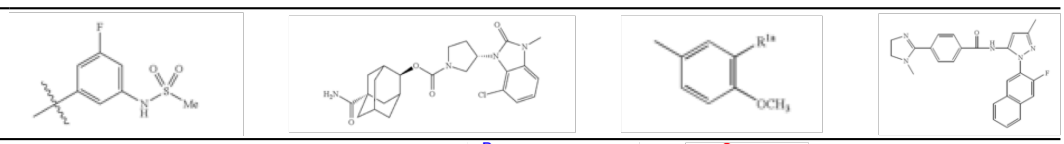
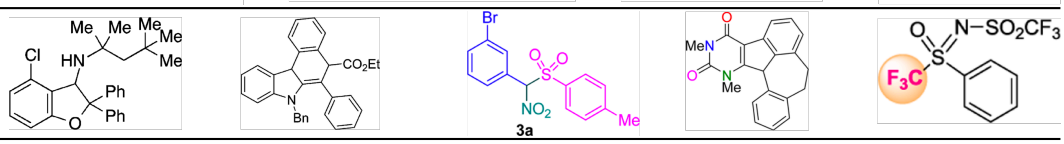
Dataset	Example Images
Indigo	
ChemDraw	
CLEF	
JPO	
UOB	
USPTO	
Staker	
ACS	

Figure 2: Example molecular images in the test sets. The first two rows are synthetic images rendered by Indigo and ChemDraw. CLEF, JPO, UOB, USPTO, Staker, and ACS are datasets with realistic images.

and are available at `https://github.com/bayer-science-for-a-better-life/Img2Mol/`. We additionally create a new benchmark ACS with molecular images from the chemistry literature. We randomly sample molecular images cropped from chemistry publications and ask chemists to manually label their SMILES.

Figure 2 shows example images in our benchmarks. The processed datasets can be downloaded at `https://github.com/thomas0809/MolScribe#data`.

Data Augmentation

Molecule Augmentation Indigo can automatically render images of molecules, but never uses abbreviations or R-groups. In order for the synthetic images to cover more molecular patterns, we design the following molecule augmentation strategies:

- Abbreviation: We compile a list of 53 common functional group substitution rules, and randomly replace functional groups with their abbreviations in the training data. It is implemented by the substructure matching function in Indigo. The full substitution list can be found in our codebase.
- R-group: We randomly add R-group atoms to the training molecules. The R-group label is randomly sampled from the list $[\mathrm{R}, \mathrm{R}_1, \mathrm{R}_2, \ldots, \mathrm{R}_{12}, \mathrm{R}_\mathrm{a}, \mathrm{R}_\mathrm{b}, \mathrm{R}_\mathrm{c}, \mathrm{R}_\mathrm{d}, \mathrm{X}, \mathrm{Y}, \mathrm{Z}, \mathrm{A}, \mathrm{Ar}]$.
- Aromaticity: We randomly choose the way aromatic rings are represented: a circle or alternating single and double bonds.
- Explicit Hydrogens: We randomly add hydrogens as explicit atoms.

Image Augmentation The synthetic images are dynamically rendered by Indigo during training with the following options:

- render-background-color: 1,1,1;
- render-relative-thickness: randomly sampled from $[0.5, 2]$;
- render-bond-line-width: randomly sampled from $[1, 3.5]$;
- render-font-family: randomly chosen from {Arial, Times, Courier, Helvetica};
- render-label-mode: randomly chosen from {hetero, terminal-hetero};
- render-implicit-hydrogens-visible: randomly chosen from {true, false}.

We also apply image augmentation during training with the following operations applied to the image:

- rotate by a random angle chosen from $[-90°, 90°]$;
- crop each side of the image by at most 1%;
- pad one side of the image by at most 40%;
- downscale the image by 20–50% and upscale back;
- blur the image using a randomly-sized kernel;
- add Gaussian noise to the image;
- add salt-and-pepper noise (random black pixels) to the image.

The image augmentations are implemented with Albumentations.[6]

The rendering options and augmentation operations are chosen such that the augmented images have different styles and qualities, and preserve the necessary information for a human to understand the molecule structure.

Expansion of Abbreviated Structures

The algorithm for expanding abbreviated structures presented in the main manuscript makes three assumptions: 1) the structure of the functional group consists of a single-chain backbone to which side atoms are attached; 2) the atoms in the formula are listed in order along the chain; and 3) every atom only has one possible valence. (For example, for a backbone XY with side atoms A attached to X and B, C attached to Y, we assume the formula can be written XAYBC or XAYCB.) In the full algorithm, we further address the following cases:

1. *Carbon chains.* A chain of carbons with homogeneous side atoms is often written C_aX_b instead of $CX_kCX_k\ldots$, where X is usually hydrogen or a halogen. This issue is

handled by splitting any occurrence of C_aX_b to $CX_q \ldots CX_qCX_{q+r}$ where q and r are the quotient and remainder upon dividing b by a.

2. *Abbreviations.* The formulas often have abbreviations within them (e.g. Me, Et). They are treated as superatoms with an equivalent valence. In the end, they are expanded to recover their original graph structure.

3. *Nested formulas.* The formulas can be nested (e.g. $N(CH_3)_2$). Similar to abbreviations, we treat nested formulas as superatoms. We assume they are attached to the current backbone with a single bond. We then recursively apply our algorithm to the nested formula to obtain its structure and attach it to the current atom (a_{cur} in Algorithm 1 in the main manuscript). If the superatom does not saturate the valence of the current atom, we additionally require that the superatom is a side atom (i.e., not part of the backbone). If this cannot be achieved (i.e., the superatom's implicit valence is not zero after expanding the nested formula), we return FAILURE. On the other hand, if the current atom's valence becomes saturated, then the superatom joins the backbone and becomes the new a_{cur}.

4. *Multiple valences.* Some atoms have multiple possible valences. To handle this, we try each possible valence until we succeed. We try common valences before rare ones and higher valences before lower ones.

Additional Results

Table 1 shows the evaluation results when chirality is ignored. Our main evaluation metric, i.e. the exact matching accuracy as described in the paper, requires chirality to be correctly predicted. It can raise concerns of unfair comparison with Img2Mol, which does not predict chirality. So we run an additional evaluation that ignores chirality. The gap between the baseline and MolScribe becomes smaller under this metric, and sometimes the baseline even

Table 1: Accuracy of the predicted molecules when chirality is ignored (scores are in %).

	Synthetic		Realistic						Perturbed			
	Indigo	ChemDraw	CLEF	JPO	UOB	USPTO	Staker	ACS	CLEF_p	UOB_p	USPTO_p	Staker_p
MolVec	95.5	89.0	84.1	68.9	80.8	91.4	0.8	49.9	44.7	74.7	30.8	5.3
OSRA	95.0	88.1	88.5	55.8	78.7	89.1	0.0	58.6	11.6	68.5	4.1	5.1
Img2Mol	69.4	54.1	34.3	16.4	68.7	30.0	20.1	24.5	42.1	74.9	34.2	51.7
DECIMER	75.2	90.6	64.9	52.7	**88.5**	44.6	50.6	48.0	73.1	**87.7**	50.3	80.4
SwinOCSR	77.9	81.1	32.1	14.0	45.0	28.4	–	29.0	33.8	–	–	–
Baseline	96.3	**95.2**	91.4	72.4	88.4	**95.3**	95.6	64.1	93.8	87.3	**95.0**	**91.9**
MolScribe	**98.1**	93.9	**91.9**	**77.1**	88.2	94.5	95.4	**76.1**	**94.2**	86.9	93.9	87.9

Table 2: Tanimoto similarity based on Morgan fingerprints between the prediction and the ground truth (scores are in %).

	Synthetic		Realistic						Perturbed			
	Indigo	ChemDraw	CLEF	JPO	UOB	USPTO	Staker	ACS	CLEF_p	UOB_p	USPTO_p	Staker_p
MolVec	98.6	95.9	88.5	85.0	91.5	95.8	23.8	66.7	80.0	88.2	69.1	36.5
OSRA	98.1	94.8	88.9	70.5	90.4	94.1	17.4	63.0	54.4	85.0	43.2	34.1
Img2Mol	91.0	84.3	74.1	48.5	84.7	72.1	63.0	49.6	79.8	88.6	75.5	79.9
DECIMER	92.9	97.6	80.3	77.3	**95.7**	80.1	72.7	60.2	91.0	95.5	82.5	94.7
SwinOCSR	89.3	90.9	59.4	49.3	90.4	65.8	–	41.7	61.1	–	–	–
Baseline	98.6	**98.7**	89.9	**89.3**	95.8	**97.7**	**97.8**	74.6	98.0	95.4	**99.2**	**97.6**
MolScribe	**99.0**	97.5	**90.5**	88.1	**96.0**	97.5	**97.8**	**76.8**	**98.2**	**95.5**	99.1	95.5

performs better. As the chirality is ignored in this metric, the strength of MolScribe can be concealed.

Table 2 shows another evaluation metric, the Tanimoto similarity based on Morgan fingerprints, which is also reported in some of the previous works. It is a relaxed evaluation metric compared to the exact matching accuracy, as the predictions that are structurally similar to the ground truth but only differ by a few atoms will also be assigned a high similarity score.

Figure 3 shows some examples of MolScribe's prediction on hand-drawn molecular images from the DECIMER-HDM[7] dataset. Although we do not seek to handle hand-drawn images in the current model, we observe MolScribe is able to correctly recognize some of these images. Among its mistakes, many of them only differ with the ground truth by a few atoms or bonds. The overall exact matching accuracy on this dataset is 11.2%, which still has a large space for improvement. Our framework can be extended to cover hand-drawn molecules

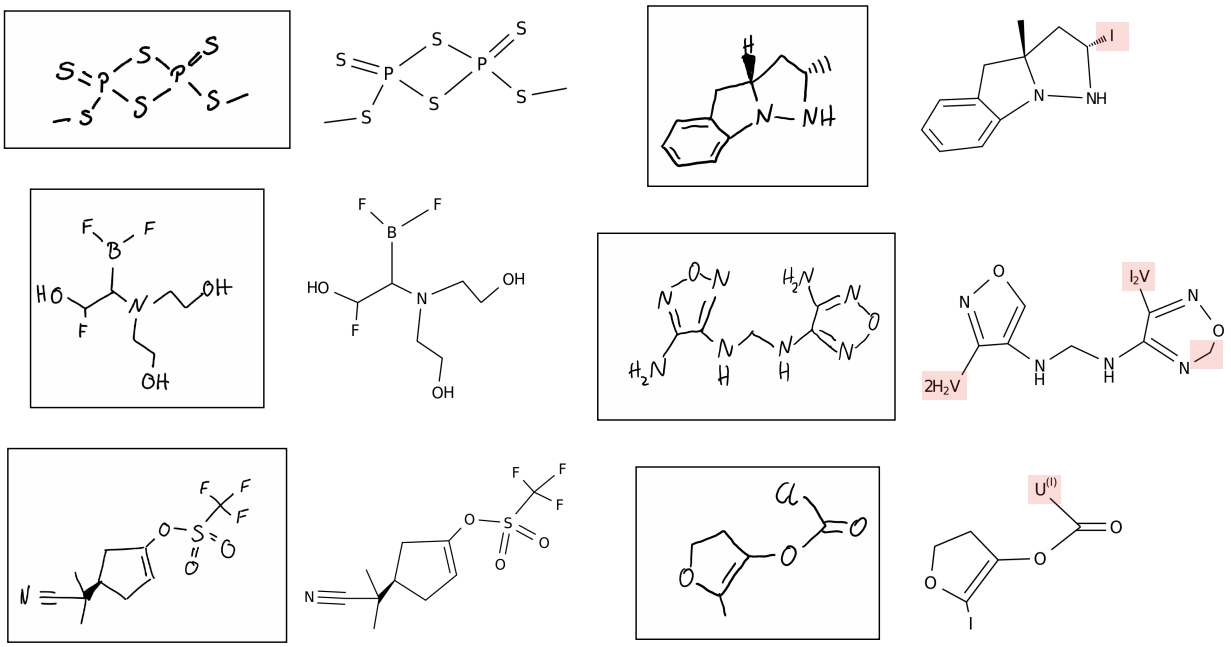


Figure 3: Examples of hand-drawn molecular images from the DECIMER-HDM dataset and MolScribe's prediction. The left three predictions are correct and the right three are wrong. Errors are indicated in red shade.

by designing data augmentation methods that synthesize hand-drawn style images.

References

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